Solutions to 2009 Power Question

1. (a) $(312, 123, 231)$

   (b) There are three: $41352, 42351, 51342$.

   (c) There are four: $352146, 362145, 452136, 562134$.

2. (a) We will prove this by contradiction. Suppose for some other 4-label L we have $S_3[L] = S_3[1234] = (123, 123)$. Write out L as a_1, a_2, a_3, a_4. From the first window of $S_3[L]$, we have $a_1 < a_2 < a_3$. From the second window, we have $a_2 < a_3 < a_4$. Connecting these inequalities gives $a_1 < a_2 < a_3 < a_4$, which forces $L = 1234$, a contradiction. Therefore, the 3-signature above is unique.

   (b) There are 11 others (12 in all, if we include $S_3[1234]$):

$$\begin{array}{lll} S_3[1234] = (123, 123) & S_3[1243] = (123, 132) & S_3[1324] = (132, 213) \\ S_3[1423] = (132, 213) & S_3[2134] = (213, 123) & S_3[2314] = (231, 213) \\ S_3[3241] = (213, 231) & S_3[3421] = (231, 321) & S_3[4132] = (312, 132) \\ S_3[4231] = (312, 231) & S_3[4312] = (321, 312) & S_3[4321] = (321, 321) \end{array}$$

   (c) If $S_3[a_1, a_2, a_3, a_4] = (123, 321)$, then the first window forces $a_2 < a_3$, whereas the second window forces $a_2 > a_3$. This is impossible, so the 3-signature $(123, 321)$ is impossible.

   (d) There are 18 impossible 3-signatures with two windows. In nine of these, the first window indicates that $a_2 < a_3$ (an increase), but the second window indicates that $a_2 > a_3$ (a decrease). In the other nine, the end of the first window indicates a decrease (that is, $a_2 > a_3$), but the beginning of the second window indicates an increase ($a_2 < a_3$). In general, for a 3-signature to be possible, the end of the first window and beginning of the second window must be consistent, indicating either an increase or a decrease. The impossible 3-signatures are:

$$\begin{array}{llllll} (123, 321) & (123, 312) & (123, 213) & (132, 231) & (132, 132) & (132, 123) \\ (213, 321) & (213, 312) & (213, 213) & (231, 231) & (231, 132) & (231, 123) \\ (312, 321) & (312, 312) & (312, 213) & (321, 231) & (321, 132) & (321, 123) \end{array}$$

3. (a) The first pair indicates an increase; the next three are decreases, and the last pair is an increase. So the 2-signature is $(12, 21, 21, 21, 12)$.

   (b) There are several:

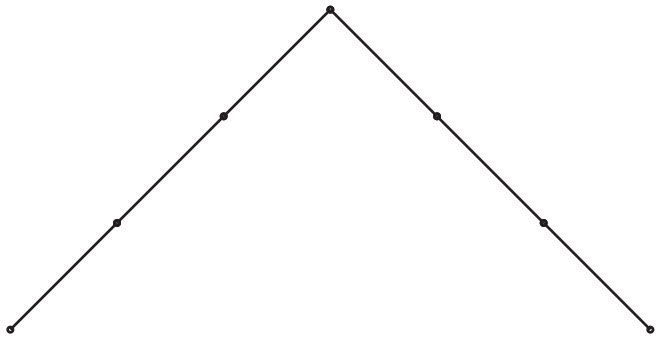
$$\begin{array}{cccccc} 564312 & 564213 & 563214 & 465312 & 465213 & 463215 \\ 365412 & 365214 & 364215 & 265413 & 265314 & 264315 \\ & & & 165413 & 165314 & 164315 \\ & & 453216 & 354216 & 254316 & 154326 \end{array}$$

4. In part (a), we can count by brute force, or use the formula from part (b) (with independent proof).

   (a) For the case of 5-labels, brute force counting is tractable.

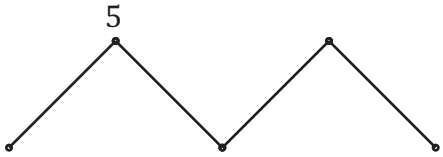
   $$\boxed{12543, 13542, 14532, 23541, 24531, 34521}.$$

   (b) The answer is $\boxed{\binom{2n}{n}}$. The shape of this signature is a wedge: n up steps followed by n down steps. The wedge for $n = 3$ is illustrated below:

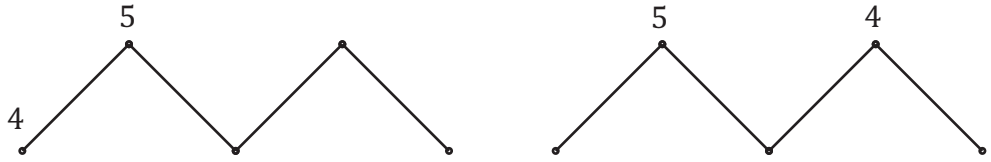
   

   The largest number in the label, $2n + 1$, must be placed at the peak in the center. If we choose the numbers to put in the first n spaces, then they must be placed in increasing order. Likewise, the remaining n numbers must be placed in decreasing order on the downward sloping piece of the shape. Thus there are exactly $\binom{2n}{n}$ such labels.

5. (a) The answer is $\boxed{16}$. We have a shape with two peaks and a valley in the middle. The 5 must go on one of the two peaks, so we place it on the first peak. By the shape's symmetry, we will double our answer at the end to account for the 5-labels where the 5 is on the other peak.

   The 4 can go to the left of the 5 or at the other peak. In the first case, shown below left, the 3 must go at the other peak and the 1 and 2 can go in either order. In the latter case, shown below right, the $1, 2$, and 3 can go in any of 3! arrangements.

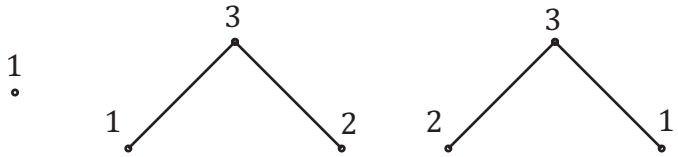
   

So there are $2! + 3! = 8$ possibilities. In all, there are 16 5-labels (including the ones where the 5 is at the other peak).

(b) The answer is $\boxed{7936}$. The shape of this 2-signature has four peaks and three intermediate valleys:



We will solve this problem by building up from smaller examples. Let f_n equal the number of $(2n+1)$-labels whose 2-signature consists of n peaks and $n-1$ intermediate valleys. In part (b) we showed that $f_2 = 16$. In the case where we have one peak, $f_1 = 2$. For the trivial case (no peaks), we get $f_0 = 1$. These cases are shown below.



Suppose we know the peak on which the largest number, $2n+1$, is placed. Then that splits our picture into two shapes with fewer peaks. Once we choose which numbers from $1, 2, \ldots, 2n$ to place each shape, we can compute the number of arrangements of the numbers on each shape, and then take the product. For example, if we place the 9 at the second peak, as shown below, we get a 1-peak shape on the left and a 2-peak shape on the right.



For the above shape, there are $\binom{8}{3}$ ways to pick the three numbers to place on the left-hand side, $f_1 = 2$ ways to place them, and $f_2 = 16$ ways to place the remaining five numbers on the right.

This argument works for any $n > 1$, so we have shown the following:

$$f_n = \sum_{k=1}^{n} \binom{2n}{2k-1} f_{k-1} f_{n-k}.$$

So we have:

$$f_1 = \binom{2}{1} f_0^2 = 2$$
$$f_2 = \binom{4}{1} f_0 f_1 + \binom{4}{3} f_1 f_0 = 16$$
$$f_3 = \binom{6}{1} f_0 f_2 + \binom{6}{3} f_1^2 + \binom{6}{5} f_2 f_0 = 272$$
$$f_4 = \binom{8}{1} f_0 f_3 + \binom{8}{3} f_1 f_2 + \binom{8}{5} f_2 f_1 + \binom{8}{7} f_3 f_0 = \boxed{7936}.$$

6. (a) Signature (i) is possible, because it is the 3-signature of 12435.
Signature (ii) is impossible. Let a 5-label be a_1, a_2, a_3, a_4, a_5. The second window of (ii) implies $a_3 < a_4$, whereas the third window implies $a_3 > a_4$, a contradiction.

   (b) There are $\boxed{p}$ such windows, regardless of ω_1.

   (c) Because the windows ω_1 and ω_2 overlap for $p-1$ numbers in L, the first $p-1$ integers in ω_2 must be in the same relative order as the last $p-1$ integers in ω_1. Therefore, by choosing the last integer in ω_2 (there are p choices), the placement of the remaining $p-1$ integers is determined. We only need to show that there exists some L such that $S_p[L] = (\omega_1, \omega_2)$.

   To do so, set $\omega_1 = w_1, w_2, \ldots, w_p$. Provisionally, let $L^{(k)} = w_1, w_2, \ldots, w_p, k + 0.5$ for $k = 0, 1, \ldots, p$. For example, if $p = 4$ and $\omega_1 = 3124$, then $L^{(2)} = 3, 1, 2, 4, 2.5$. First we show that $L^{(k)}$ has the required p-signature; then we can renumber the entries in $L^{(k)}$ in consecutive order to make them all integers, so for example $3, 1, 2, 4, 2.5$ would become $4, 1, 2, 5, 3$.

   Even though the last entry in $L^{(k)}$ is not an integer, we can still compute $S_p[L^{(k)}]$, since all we require is that the entries are all distinct. In each $S_p[L^{(k)}]$, the first window is ω_1. When we compare $L^{(k)}$ to $L^{(k+1)}$, the last integer in ω_2 can increase by at most 1, because the last integer in the label jumps over at most one integer in positions 2 through p (those jumps are in boldface):

$$\begin{aligned}
L^{(0)} &= 3, 1, 2, 4, 0.5 & S_4[L^{(0)}] &= (3124, 234\underline{1}) & \\
L^{(1)} &= 3, \mathbf{1}, 2, 4, 1.5 & S_4[L^{(1)}] &= (3124, 134\underline{2}) & \text{[jumps over the 1]} \\
L^{(2)} &= 3, 1, \mathbf{2}, 4, 2.5 & S_4[L^{(2)}] &= (3124, 124\underline{3}) & \text{[jumps over the 2]} \\
L^{(3)} &= 3, 1, 2, 4, 3.5 & S_4[L^{(3)}] &= (3124, 124\underline{3}) & \text{[jumps over nothing]} \\
L^{(4)} &= 3, 1, 2, \mathbf{4}, 4.5 & S_4[L^{(4)}] &= (3124, 123\underline{4}) & \text{[jumps over the 4]}
\end{aligned}$$

   The final entry of ω_2 in $S_p[L^{(0)}]$ is 1, because 0.5 is smaller than all other entries in $L^{(0)}$. Likewise, the final entry of ω_2 in $S_p[L^{(p)}]$ is p. Since the final entry increases in increments of 0 or 1 (as underlined above), we must see all p possibilities for ω_2.

By replacing the numbers in each $L^{(k)}$ with the integers 1 through $p+1$ (in the same relative order as the numbers in $L^{(k)}$), we have found the $(p+1)$-labels that yield all p possibilities.

7. (a) The answer is $\boxed{p! \cdot p^{n-p}}$.
Call two consecutive windows in a p-signature **compatible** if the last $p-1$ numbers in the first label and the first $p-1$ numbers in the second label (their "overlap") describe the same ordering. For example, in the p-signature $(\ldots, 2143, 2431, \ldots)$, 2143 and 2431 are compatible. Notice that the last three digits of 2143 and the first three digits of 2431 can be described by the same 3-label, 132.

Theorem: A signature σ is possible if and only if every pair of consecutive windows is compatible.

Proof: ($\Rightarrow$) Consider a signature σ describing a p-label L. If some pair in σ is not compatible, then there is some string of $p-1$ numbers in our label L that has two different $(p-1)$-signatures. This is impossible, since the p-signature is well-defined.

($\Leftarrow$) Now suppose σ is a p-signature such that that every pair of consecutive windows is compatible. We need to show that there is at least one label L with $S_p[L] = \sigma$. We do so by induction on the number of windows in σ, using the results from 5(b).

Let $\sigma = \{\omega_1, \omega_2, \ldots, \omega_{k+1}\}$, and suppose $\omega_1 = a_1, a_2, \ldots, a_p$. Set $L_1 = \omega_1$.

Suppose that L_k is a $(p+k-1)$-label such that $S_p[L_k] = \{\omega_1, \ldots, \omega_k\}$. We will construct L_{k+1} for which $S_p[L_{k+1}] = \{\omega_1, \ldots, \omega_{k+1}\}$.

As in 5(b), denote by $L_k^{(j)}$ the label L_k with a $j + 0.5$ appended; we will eventually renumber the elements in the label to make them all integers. Appending $j + 0.5$ does not affect any of the non-terminal windows of $S_p[L_k]$, and as j varies from 0 to $p-k+1$ the final window of $S_p[L_k^{(j)}]$ varies over each of the p windows compatible with ω_k. Since ω_{k+1} is compatible with ω_k, there exists some j for which $S_p[L_k^{(j)}] = \{\omega_1, \ldots, \omega_{k+1}\}$. Now we renumber as follows: set $L_{k+1} = S_{k+p}[L_k^{(j)}]$, which replaces $L_k^{(j)}$ with the integers 1 through $k+p$ and preserves the relative order of all integers in the label.

By continuing this process, we conclude that the n-label L_{n-p+1} has p-signature σ, so σ is possible. □

To count the number of possible p-signatures, we choose the first window ($p!$ choices), then choose each of the remaining $n-p$ compatible windows (p choices each). In all, there are $p! \cdot p^{n-p}$ possible p-signatures.

(b) The answer is $\boxed{n = 7, p = 5}$.
Let P denote the probability that a randomly chosen p-signature is possible. We are

given that $1 - P = 575$, so $P = \frac{1}{576}$. We want to find p and n for which

$$\begin{aligned} \frac{p! \cdot p^{n-p}}{(p!)^{n-p+1}} &= \frac{1}{576} \\ \frac{p^{n-p}}{(p!)^{n-p}} &= \frac{1}{576} \\ ((p-1)!)^{n-p} &= 576. \end{aligned}$$

The only factorial that has 576 as an integer power is $4! = \sqrt{576}$. Thus $p = 5$ and $n - p = 2 \Rightarrow n = 7$.

8. (a) The p-signature is not unique because it equals both $S_3[625143]$ and $S_3[635142]$.

   (b) Let $L = a_1, a_2, a_3, a_4, a_5, a_6$. We have $a_4 < a_6 < a_5$ (from window #4), $a_3 < a_1 < a_2$ (from window #1), and $a_2 < a_4$ (from window #2). Linking these inequalities, we get

   $$a_3 < a_1 < a_2 < a_4 < a_6 < a_5 \quad \Rightarrow \quad L = 231465,$$

   so $S_3[L]$ is unique.

   (c) 12345 and 54321 are the only ones.

   (d) $L = 645132$ will work. First, note that $S_3[645132] = (312, 231, 312, 132)$. Next, we need to show that $S_4[645132] = \{4231, 3412, 4132\}$ is unique. So let $L' = a_1, a_2, a_3, a_4, a_5, a_6$ be a 6-label such that $S_4[L'] = (4231, 3412, 4132)$.

   We get $a_4 < a_6 < a_5$ (from window #3), $a_5 < a_2 < a_3$ (from window #2), and $a_3 < a_1$ (from window #1). Linking these inequalities, we get

   $$a_4 < a_6 < a_5 < a_2 < a_3 < a_1 \quad \Rightarrow \quad L' = 645132,$$

   so $L' = L$, which means $S_4[L]$ is unique.

9. (a) The n-labels with unique 2-signatures are $1, 2, \ldots, n$ and $n, n-1, \ldots, 1$, and their respective 2-signatures are $(12, 12, \ldots, 12)$ and $(21, 21, \ldots, 21)$.

   Proof: (Not required for credit.) Let $L = a_1, a_2, \ldots, a_n$. The first signature above implies that $a_1 < a_2 < \cdots < a_n$, which forces $a_1 = 1, a_2 = 2$, and so on. Likewise, the second signature forces $a_1 = n, a_2 = n - 1$, and so on.

   To show that all other n-labels fail to have unique 2-signatures, we will show that in any other n-label L', there are two numbers k and $k+1$ that are not adjacent. By switching k and $k+1$ we get a label L'' for which $S_2[L'] = S_2[L'']$, since the differences between k and its neighbors in L' were at least 2 (and likewise for $k+1$).

   To show that such a k and $k+1$ exist, we proceed by contradiction. Suppose that all such pairs are adjacent in L'. Then 1 and n must be at the ends of L' (or else some intermediate number k will fail to be adjacent to $k+1$ or $k-1$). But if $a_1 = 1$, then

that forces $a_2 = 2$, $a_3 = 3, \ldots, a_n = n$. That is, we get the two labels we already covered. (We get the other label if $a_n = 1$.)

Therefore, none of the $n! - 2$ remaining labels has a unique 2-signature.

(b) $S_5[495138627]$ is unique.
Let $L = a_1, \ldots, a_9$ and suppose $S_5[L] = S_5[495138627] = (\omega_1, \ldots, \omega_5)$. Then we get the following inequalities:

$a_4 < a_8$	[from ω_4]	$a_8 < a_5$	[from ω_4]
$a_5 < a_1$	[from ω_1]	$a_1 < a_3$	[from ω_1]
$a_3 < a_7$	[from ω_3]	$a_7 < a_9$	[from ω_5]
$a_9 < a_6$	[from ω_5]	$a_6 < a_2$	[from ω_2]

Combining, we get $a_4 < a_8 < a_5 < a_1 < a_3 < a_7 < a_9 < a_6 < a_2$, which forces $a_4 = 1, a_8 = 2, \ldots, a_2 = 9$. So the label L is forced and $S_5[495138627]$ is therefore unique.

(c) The answer is $\boxed{p = 16}$. To show this fact we will need to extend the idea from part 8(b) about "linking" inequalities forced by the various windows:

Theorem: A p-signature for an n-label L is unique if and only if for every $k < n$, k and $k + 1$ are in at least one window together. That is, the **distance** between them in the n-label is less than p.

Proof: Suppose that for some k, the distance between k and $k+1$ is p or greater. Then the label L' obtained by swapping k and $k + 1$ has the same p-signature, because there are no numbers between k and $k+1$ in any window and because the two numbers never appear in the same window.

If the distance between all such pairs is less than k, we need to show that $S_p[L]$ is unique. For $i = 1, 2, \ldots, n$, let r_i denote the position where i appears in L. For example, if $L = 4123$, then $r_1 = 2, r_2 = 3, r_3 = 4$, and $r_4 = 1$.

Let $L = a_1, a_2, \ldots, a_n$. Since 1 and 2 are in some window together, $a_{r_1} < a_{r_2}$. Similarly, for any k, since k and $k + 1$ are in some window together, $a_{r_k} < a_{r_{k+1}}$. We then get a linked inequality $a_{r_1} < a_{r_2} < \cdots < a_{r_n}$, which can only be satisfied if $a_{r_1} = 1, a_{r_2} = 2, \ldots, a_{r_n} = n$. Therefore, $S_p[L]$ is unique. □

From the proof above, we know that the signature is unique if and only if every pair of consecutive integers coexists in at least one window. Therefore, we seek the largest distance between consecutive integers in L. That distance is 15 (from 8 to 9, and from 17 to 18). Thus the smallest p is $\boxed{16}$.

10. Let s_k denote the number of such unique signatures. We proceed by induction with base case $k = 2$. From 8(c), a 2-signature for a label L is unique if and only if consecutive numbers in L appear together in some window. Because $k = 2$, the consecutive numbers must be adjacent

in the label. The 2^2-labels 1234 and 4321 satisfy this condition,[1] so their 2^1-signatures are unique. Thus we have shown that $s_2 \geq 2 = 2^{2^2-3}$, and the base case is established.

Now suppose $s_k \geq 2^{2^k-3}$ for some $k \geq 2$. Let L_k be a 2^k-label with a unique 2^{k-1}-signature. Write $L_k = (a_1, a_2, \ldots, a_{2^k})$. We will *expand* L_k to form a 2^{k+1} label by replacing each a_i above with the numbers $2a_i - 1$ and $2a_i$ (in some order). This process produces a valid 2^{k+1}-label, because the numbers produced are all the integers from 1 to 2^{k+1}. Furthermore, different L_k's will produce different labels: if the starting labels differ at place i, then the new labels will differ at places $2i - 1$ and $2i$. Therefore, each starting label produces 2^{2^k} distinct 2^{k+1}-labels through this process. Summarizing, each valid 2^k-label can be expanded to produce 2^{2^k} distinct 2^{k+1}-labels, none of which could be obtained by expanding any other 2^k-label.

It remains to be shown that the new label has a unique 2^{k-1}-signature. Because L_k has a unique 2^{k-1}-signature, for all $i \leq 2^k - 1$, both i and $i + 1$ appeared in some 2^{k-1}-window. Therefore, there were fewer than $2^{k-1} - 1$ numbers between i and $i + 1$. When the label is expanded, $2i$ and $2i - 1$ are adjacent, $2i + 1$ and $2i + 2$ are adjacent, and $2i$ and $2i + 1$ are fewer than $2 \cdot (2^{k-1} - 1) + 2 = 2^k$ places apart. Thus, every pair of adjacent integers is within some 2^k-window.

Since each pair of consecutive integers in our new 2^{k+1}-label coexists in some 2^k-window for every possible such expansion of L_k, that means all 2^{2^k} ways of expanding L_k to a 2^{k+1}-label result in labels with unique 2^k-signatures. We then get

$$\begin{aligned} s_{k+1} &\geq 2^{2^k} \cdot s_k \\ &\geq 2^{2^k} \cdot 2^{2^k-3} \\ &= 2^{2^{k+1}-3}, \end{aligned}$$

which completes the induction.

[1]The proof in 8(a) establishes that these are the only 2^2-labels with unique 2-signatures, thus giving $s_2 = 2$. That proof was not required for credit and is not needed here, since the inequality above is good enough for this problem.